250–500

This 15th-century painting depicts the seven liberal arts, core subjects such as arithmetic, music, astronomy, rhetoric, and grammar that the 5th-century writer Martianus Capella established as the basis of early medieval European education.

DIOPHANTUS OF ALEXANDRIA (*c.*200–*c.*284) founded the mathematical discipline of algebra around 250 by introducing a systematic notation to indicate an unknown quantity and its power; for example, in the equation $x^2 - 3 = 6$, x^2 represents an unknown number raised to the power of 2 (or squared). In his *Arithmetica*, Diophantus provided solutions for linear equations (in which no variable in the equation is raised to a power greater than 1—as in $ax + b = 0$) and quadratic equations (in which at least one of the variables is squared—as in $ax^2 + bx + c = 0$). Diophantus also made a particular study of **indeterminate equations**, proposing a method of solving them that is now known as Diophantine analysis. Fermat's Last Theorem (see 1635–37) is probably the most famous example of such an equation.

Roman surgical instruments
Ancient Roman physicians used a wide variety of surgical instruments, including spatulas and hooks (right), specula for internal examinations, and saws.

In around 320, **Pappus of Alexandria** (*c.*290–*c.*350) compiled *Collections*, an eight-volume work that contained the major results of the great mathematicians who preceded him and also introduced novel concepts. Among these new ideas were work on the **centers of gravity** and the volumes created by plane figures revolving. He also proposed what is now known as **Pappus's hexagon theorem**, which states that the intersections of three collinear points (points along the same line) with three similar points along a similar line will themselves be collinear.

> **❝** I HAVE ATTEMPTED TO EXPLAIN THE **NATURE** AND **POWER OF NUMBERS** BY STARTING WITH THE **FOUNDATION** ON WHICH ALL **THINGS ARE BUILT. ❞**
>
> **Diophantus of Alexandria, Greek mathematician,** from *Arithmetica, c.*250

In the 3rd century, Plotinus (*c.*205–270) created a modified form of Plato's teachings (see 700–400 BCE) known as **Neoplatonism**, which remained influential into the Middle Ages. Plotinus taught that there is a transcendent being (the "One"), which cannot be described, from which emanated a series of other beings. These included the "Divine Mind" and the "World Soul," from which human souls are derived. Plotinus's follower Iamblichus of Apamea (*c.*245–*c.*325) developed these ideas, adding number symbolism derived from Pythagoras (see 700–400 BCE). Iamblichus believed that **mathematical theorems applied to the whole universe**, including divine beings, and that numbers themselves had a form of concrete existence.

In line with the general trend in the 3rd and 4th centuries for gathering together the work of earlier scientists, **Oribasius of Pergamum** (*c.*323–400) produced *Collections*, a set of 70 volumes that **brought together the works of Galen and other earlier medical writers**. Only 20 of these volumes survive, of which four, collectively titled *Euporista*, give advice on food, drink, and diets. Oribasius also described a sling to bind a fractured jaw, which he attributes to the 1st-century physician Heraklas. Oribasius became personal physician to the Roman emperor Julian, but failed to save his patron when he was struck by a spear during a battle in Persia in 363.

In China, mathematicians continued to make advances. *Hai Tao Suan Ching*, which dates from 263, contains a discussion

Raised fields
The Maya cut drainage channels through swamps, heaping up the fertile silt to create raised fields, similar to the ones seen here.

c.250 Maya use **raised field systems**, hill terracing, and irrigation canals

c.250 Diophantus introduces **algebraic equations**

c.255–270 Plotinus develops a modified form of Plato's ideas, now known as **Neoplatonism**

263 Chinese mathematicians produce ***Hai Tao Suan Ching***

c.300 Sun Zi compiles the ***Sun Zi Suan Ching***

c.300 Iamblichus of Apamea proposes that numbers have a concrete existence and **mathematical theorems apply to the entire universe**

c.340 Pappus of Alexandria produces his work on plane figures and proposes the **hexagon theorem**